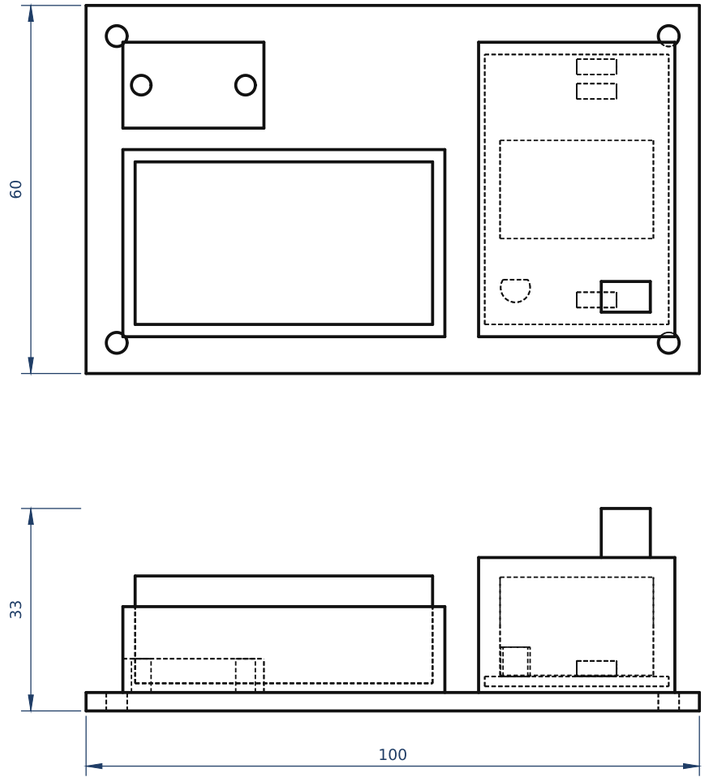
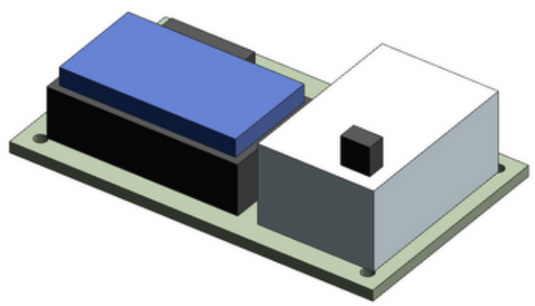
THE MODULE (its layout a placeholder [RH]) 1:1



REVISIONS		
Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	PROPOSED · the first cut (sim/parts-first-cut-findings.md); its place on the rotor OPEN
qty per machine	1
used in	the rotor (place OPEN)
source	sim/parts-first-cut-findings.md §3 (K4); sim/parts_first_cut_results.json kick

specification	value
cell	9 V lithium (LiMnO2, 1.2 Ah): Ultralife U9VL-J-P, 48.5 × 26.5 × 17.5, in a 9 V holder (Keystone 1294 class)
C kick	470 µF 25 V bipolar electrolytic, Ø16 × 25: Nichicon UES1E471MHM; it swings to −2.4 V after the dump, so it must be bipolar (or film)
R1	10 kΩ, 0.25 W: the cell charges C in τ 4.7 s
SCR	≥ 400 V, ≥ 1 A, sensitive gate: Littelfuse TCR22-8 (600 V, 1.5 A, TO-92); it sees −101 … +119 V
snubber, gate	100 Ω + 10 nF across the SCR (dV/dt); R3 1 kΩ gate to cathode; the reed through 0.1 µF + R2 100 Ω into the gate
reed, magnet	normally open, flange mount on M3: Littelfuse 59140; the magnet a Littelfuse 57140-000 actuator, brought from outside the frame
link	the arming link, a 2-pin header and jumper on the potting
energy	19.0 mJ at 9 V; the seed needs 2.2 mJ (16 % of Ψs), a fast dump about 8 mJ: × 2.4



3-D view (axonometric, not to scale)

NOTES

1 The K4 kick (PROPOSED): the cell charges C through R1; at full speed the magnet closes the reed once a revolution, the reed pulses the SCR, and the SCR dumps C across La, + to node a and − to REF: 0.60 A peak over 17 ms. The SCR stops when its dump ends, so each charge kicks once.

2 Procedure: arm at standstill (the link in); spin up to 600 rpm each way, never below 500; bring the magnet to the reed's track; the AH comes up within about 25 cycles (0.2 s); withdraw the magnet. After a slow-down below about 750 rpm relative, re-kick at full speed (C recharges in about 25 s).

3 The module's layout and envelope are placeholders [RH]; its place on the rotor is OPEN: where the cage carries no vanes, away from the bridges and utrons, the reed's blades radial and potted so 20-63 g cannot close it.

4 It starts the pump at all four rotor phases tried with every diode set (sim/parts-first-cut-findings.md §3).

DCCREG turbine · the magnetic pump: the start

Start kick K4, module		Drawing no.
Material: a G10 base, the cell in its holder, the board potted in silicone, the reed		**DCCREG-REL-730**
Mass: —	Qty per machine: 1	Rev A Sheet 61 of 103 A3
Scale: 1:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**